Empowered AI - Sense Beyond Limits

Project Team

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Chapter 1

Introduction

"Empowered-Al" project is a complex accessibility application that responds to the needs of disabled persons in an effort to improve their independence and the quality of life of the visually impaired. The app incorporates recent Al technology to facilitate communication help and interaction support, interaction helps throughout a vast number of everyday activities. With features such as computer vision, natural language processing, and haptic feedback, the "Empowered-Al" seeks to remove communication restrictions and make daily life less challenging. With reference to the above classification, features of the app are defined as the ability to detect objects in the surrounding space, receive descriptions of scenes, identify facial expressions as well as to facilitate effective communication and improve the capacity to solve routine activities in daily life. **?**.

1.1 Problem Statement

People with vision impairment experience multiple problems in mobility and management of everyday tasks. This results in making them develop a feeling of dependency, making them feel that they cannot do things on their own, thus a feeling of restricted freedom. It can be challenging to understand and interpret gestures and other nonverbal signs essential for interpersonal communication, which results in a sensation of loneliness or misinterpretation of other people's intentions and messages in social interactions.

However, the price of the existing assistive technologies can also be another challenge ahead. Some of these tools are very costly and unaffordable to a huge number of the visually impaired especially those in low-income areas. While they are still affordable, some of these devices are not adequate in offering complete solutions to execute complex tasks, as they usually focus on the particular issues that need to be solved.

Another limitation is that current assistive technologies are not fully compatible with each other, which causes the users to have to use a variety of gadgets or programs in order to fulfill their requirements during the day.

1.2 Scope

The "Empowered-AI" project is an application designed to enhance the independence of visually impaired individuals by providing essential tools for daily tasks. The app utilizes advanced AI technologies to deliver key functionalities such as real-time object detection, enabling users to identify and locate objects in their surroundings. It also produces scene descriptions through computer vision, which helps users comprehend their surroundings. Another significant aspect of this design is facial emotion recognition, indicating the definiteness of emotions and allowing the user to understand social signals. Furthermore, it has the OCR feature for the identification and scanning of printed or handwritten text within the app. Specifically, voice feedback is utilized to convey this information to users in order to create a seamless auditory experience.

This app is designed and located so that it is downloadable without cost to the users especially those in the low-income regions. In so doing, the project provides functionalities in one platform hence differing from the use of separate devices or costly assistive technologies. The Empowered-AI application needs to be created so that the visually impaired person is confident and independent with the help of the application.

Our project will not function effectively in night conditions. Although our model can detect objects similarly to the human eye at night, for clear detection and scene understanding, a night vision camera will be required, which is beyond the scope of this project.

1.3 Modules

1.3.1 Module 1: Vision Insights

This module involves working with the datasets and enhancing the ability to detect and quantify objects in different situations. It also entails perceiving different scene contexts appropriately for interaction.

1. Data Set Analysis
2. Object Detection Model Training
3. Scene Understanding

1.3.2 Module 2: Smart Interface

This module entails creating the UI and UX with Figma in addition to adding scene descriptions and text-to-speech features to improve efficiency and usability.

1. UI Figma Design

2. Scene Description

3. Text-to-Speech Integration

1.3.3 Module 3: Live Assist

This module focuses on the real action of building the application, with additional sections like text recognition or voice feedback so the user has real-time information and assistance.

1. App Development

2. Text Recognition

3. Voice Feedback

1.3.4 Module 4: Emotion Sense

This module is concerned with adding facial emotion detection technology and further testing all the functionalities of the application to ensure it runs and delivers correct outcomes.

1. Facial Emotion Detection

2. Testing and Quality Assurance

1.4 User Classes and Characteristics

User class	Description
Visually Impaired Individuals	This is the primary user group for the Empowered-AI app. These individuals rely on assistive technologies to navigate their daily lives. Users in this class may have varying degrees of vision impairment, from partial to complete blindness. They will interact with the app through voice feedback, receiving immediate audio confirmation upon selecting an option, which enhances their confidence in navigation.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Event Response Table

Event	**System State**	**Response**
User opens the app	The welcome screen is displayed, indicating readiness for interaction.	The app reads aloud, "Welcome to Empowered-AI."
User issues voice command for Object Detection	The app transitions to Object Detection mode.	The app reads aloud, "Object Detection activated. The app is now detecting objects through the camera."
User activates Scene Description	The app prepares to provide a description of the current scene.	The app reads aloud, "Scene Description activated. Please capture an image."
User issues command for Text Recognition	The app prepares to recognize printed or handwritten text.	The app reads aloud, "Text Recognition activated. The app will now read the text in front of you."
User points camera at a person	The app processes the camera input to detect the person's emotional expression.	The app reads aloud, "Emotion Detection activated. The app is analyzing the person's emotional expression."
User interacts with an object through voice command	The app receives and processes the voice command for interaction.	The app vibrates to indicate a successful interaction.
User closes the app	The app prepares to exit and complete the session.	The app reads aloud, "Thank you for using Empowered-AI. Goodbye!"

Table 2.1: User Interaction Events with System Responses

2.2 Functional Requirements

This section describes the functional requirements of the system, organized by each module and its respective features.

2.2.1 Module 1: Vision Insights

Following are the requirements for module 1:

1. FR1: The system shall be able to detect and identify objects in real-time within the user's environment.

2. FR2: The system shall provide scene descriptions based on object detection and contextual understanding.

2.2.2 Module 2: Smart Interface

Following are the requirements for module 2:

1. FR1: The system shall describe scenes in a way that visually impaired users can understand via voice feedback.

2. FR2: The system shall integrate text-to-speech technology to facilitate easy interaction.

2.2.3 Module 3: Live Assist

Following are the requirements for module 3:

1. FR1: The system shall provide real-time assistance by recognizing printed and handwritten text using OCR.

2. FR2: The system shall provide voice feedback to users based on the text detected by the app.

2.2.4 Module 4: Emotion Sense

Following are the requirements for module 4:

1. FR1: The system shall detect facial emotions in real-time using AI-based emotion recognition technology.

2. FR2: The system shall test all functionalities to ensure they provide accurate and reliable outputs.

3. FR3: The system shall conduct thorough quality assurance to ensure robustness across different environments.

2.3 Non-Functional Requirements

This section specifies non-functional requirements. These quality requirements should be specific, quantitative, and verifiable.

2.3.1 Reliability

- RELIABILITY-1: The system shall maintain a mean time between failures (MTBF) of at least 168 hours of continuous operation, with a failure defined as any unplanned interruption of service or functionality that prevents the user from completing a primary task.

- RELIABILITY-2: The application shall have a maximum allowable downtime of no more than 1 day per month for maintenance and updates, ensuring high availability for users.

2.3.2 Usability

- USABILITY-1: The application shall allow users to access the main features (object detection, scene description, text recognition, etc.) within three taps, ensuring a streamlined user experience.

- USABILITY-2: New users shall be able to learn how to use the core functionalities of the app within 30 minutes of initial use.

- USABILITY-3: The application shall provide clear and concise audio instructions for all features, minimizing the potential for user errors and ensuring effective recovery from any mistakes made during interactions.

2.3.3 Performance

- PERFORMANCE-1: 90% of scene descriptions generated by the application shall be processed and relayed to the user within 15 seconds of user input.

- PERFORMANCE-2: Voice feedback responses shall have a latency of no more than 3 seconds from the time the user requests information.

2.3.4 Security

- SECURITY-1: The system shall ensure all data is processed securely in-memory and discarded immediately after use, preventing any unauthorized access or data leaks.

- SECURITY-2: The application shall not store any personal or sensitive information locally or on external servers to maintain user privacy.

2.4 Domain Model

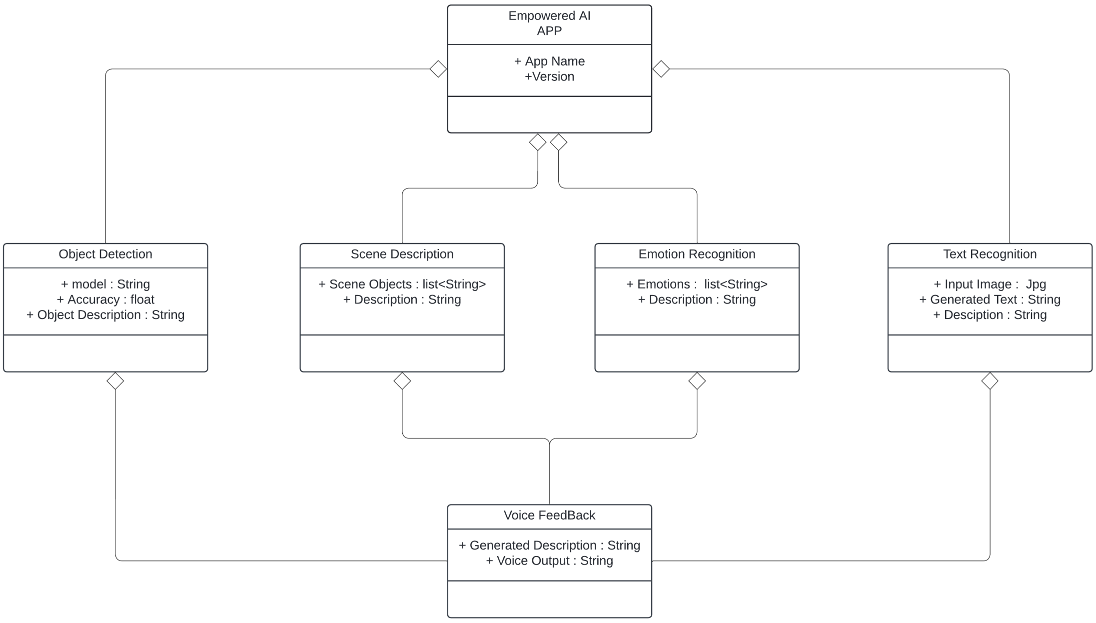

Figure 2.1: Domain Model

Chapter 3

System Overview

3.1 Architectural Design:

A **client-server architecture** fits well as it allows for the seamless delivery of complex AI functionalities while ensuring scalability and performance. The app (client) installed on users' devices can communicate with powerful servers to process intensive tasks like object detection, facial emotion recognition, and scene understanding using machine learning models. By offloading these computations to the server, The server handles the AI processing and returns real-time results, while the client provides immediate audio feedback, creating an efficient, responsive, and scalable solution for all users.

Explanation of Architecture Diagram:

In the architecture diagram, mobile clients send data from the environment, such as images or real-time streams, to a server through a network interface. The server performs tasks like object detection, text recognition, scene description, and facial emotion detection using various AI models. Once the server processes the data, the results are sent back to the mobile client. Through voice feedback represented by the ‘ loudspeaker ’ symbol, the user receives the processed data which can be scenes description, text recognition or feelings analysis. This architecture can distribute the computing burden across the server in an efficient but real time manner required for mobile applications accessibility to the visually impaired.

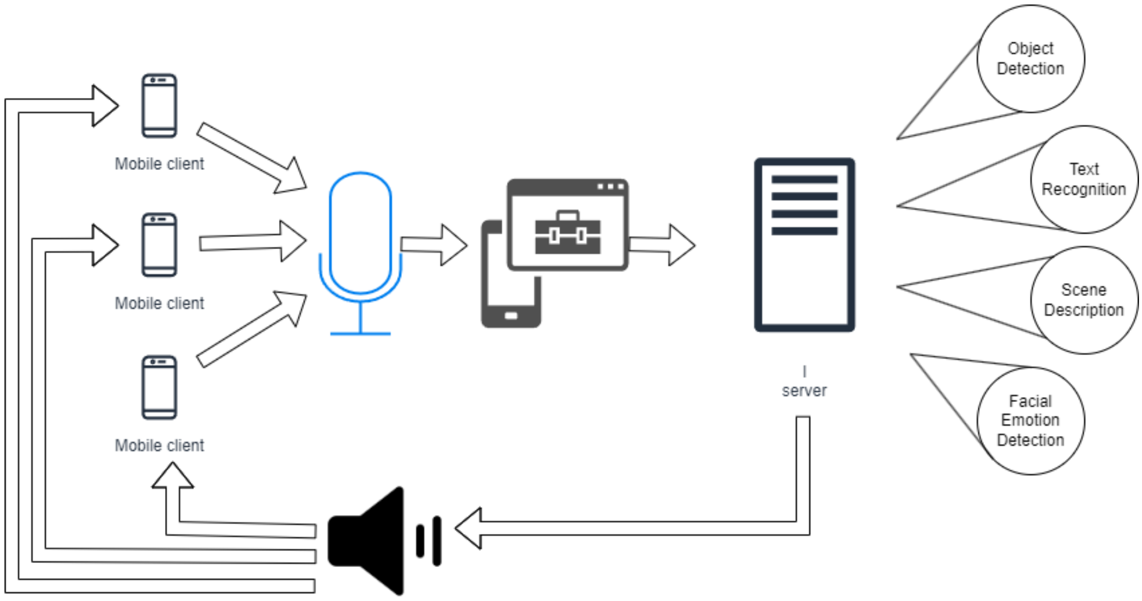

Figure 3.1: Architecture Diagram

3.2 Design Models

For this purpose, in our project, We proposed to follow Object-Oriented Development Approach (OOP) as to advantages in complexity management We identified that our system is highly complex and comprises a number of elements, including object detection, scene description, text recognition, and emotional recognition. The use of OOP we can format these components as distinct objects. This modularity helps in code accountability that is, developing, testing and experiencing each part of the system free from the influence of the other. To the credit of this concept it supports our goals of designing an extensible and sustainable application – factors that make the total development more manageable and the final product stronger.

3.2.1 Activity Diagram

3.2.1.1 Activity Diagram 1: Object Detection

The Visually impaired user launches the EmpoweredAI app and uses voice to make the application perform an object detection. It launches the camera and uses it to record a video of the environment, the video is then streamed to the server for the objects to be recognized before converting it to an audio format. It also narrates descriptions of the detected objects so that the user can have a basic idea about the environment.

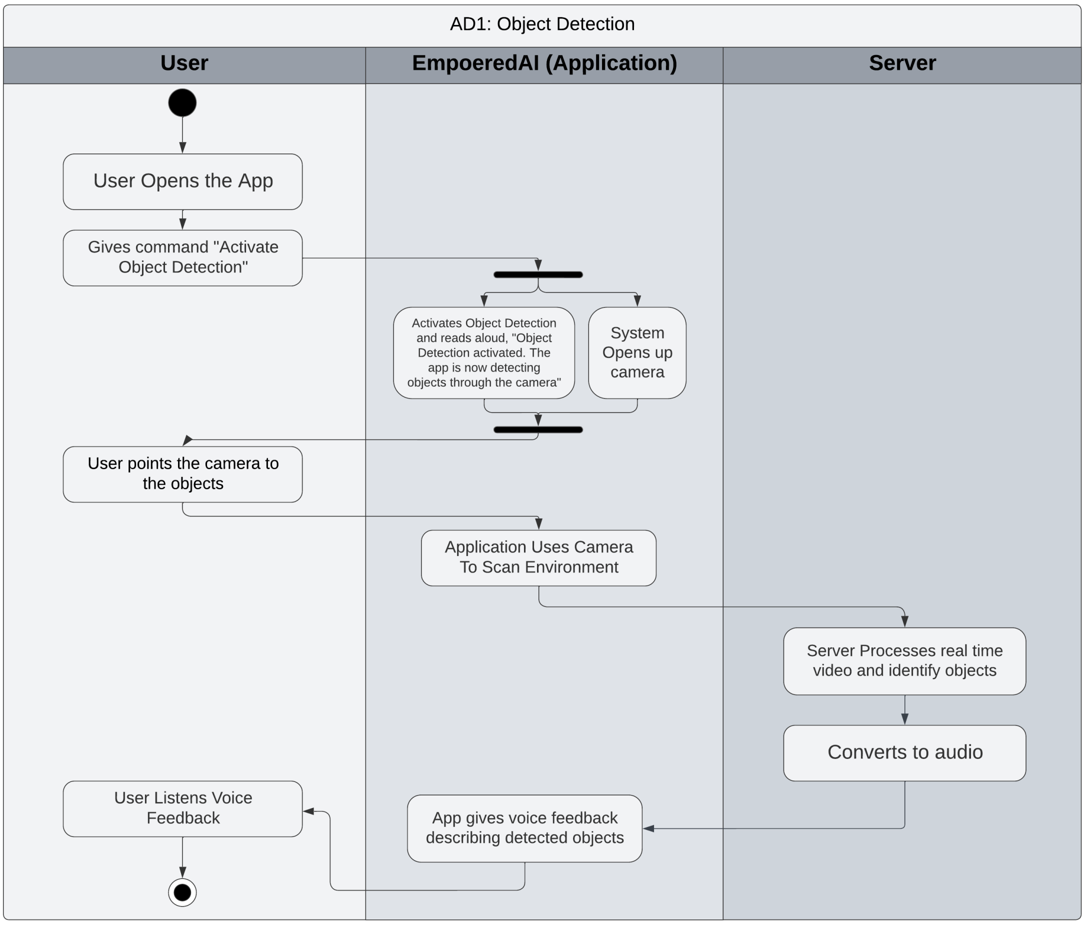

Figure 3.2: Activity Diagram 1 - Object Detection

3.2.1.2 Activity Diagram 2: Scene Description

The user launches EmpoweredAI application and utters a voice command to start the so-called "Scene Description" which asks the user to take a picture. The described image is transferred to the server, processed there by special algorithms and converted into an audio signal; the application gives an instant voiced commentary to the image for the user to listen.

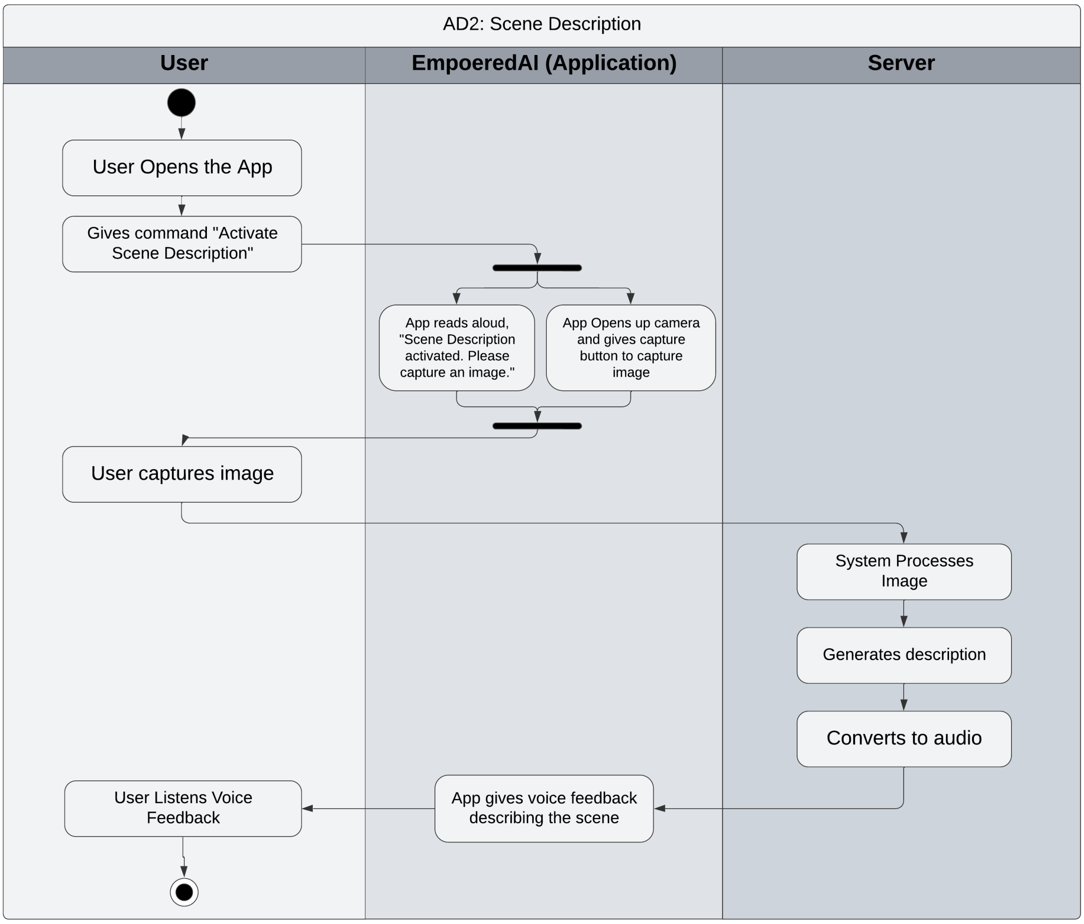

Figure 3.3: Activity Diagram 2 - Scene Description

3.2.1.3 Activity Diagram 3: Text Recognition

The user launches the Empowered AI application, and the user turns on the Text Recognition mode with a voice command as the application launches the camera. In real time, the server performs OCR to identify the text an translates it to audio so that the app can immediately provide the user with voice feedback on the pointed text.

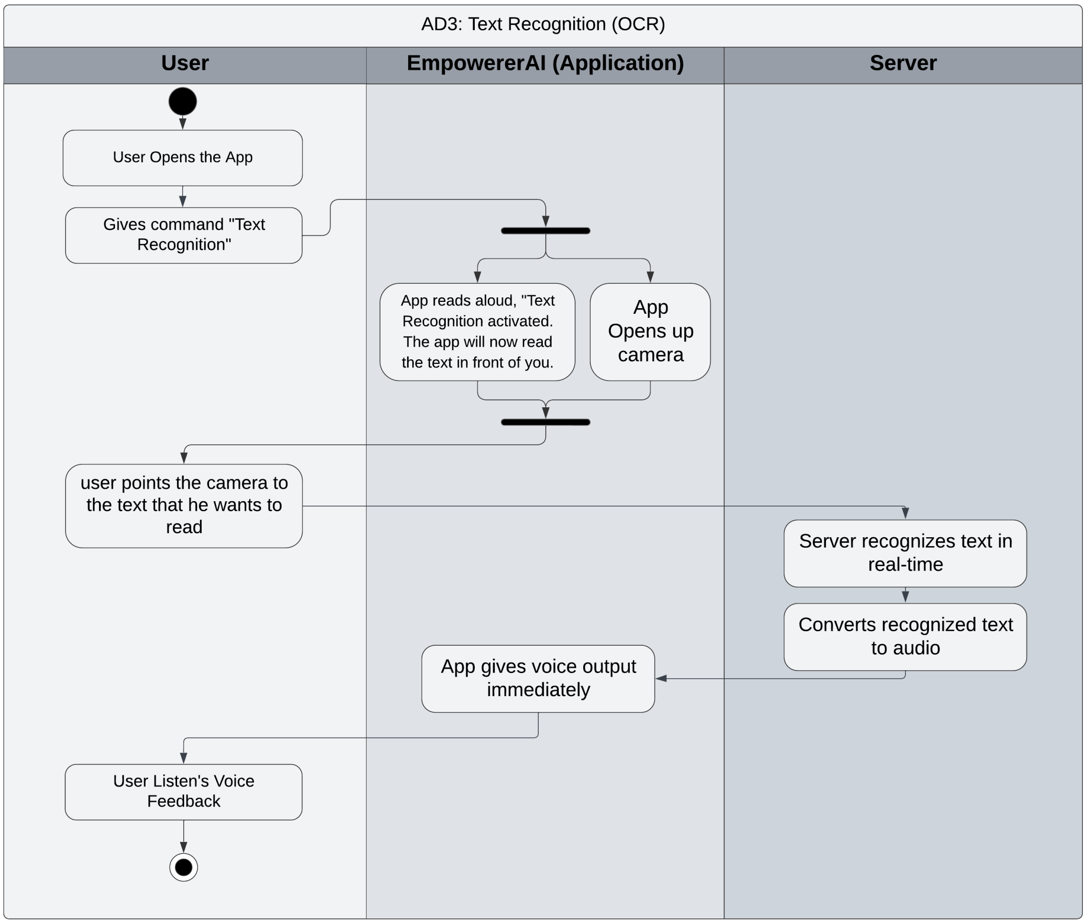

Figure 3.4: Activity Diagram 3 - Text Recognition

3.2.1.4 Activity Diagram 4: Facial Emotion Detection

The user launches the EmpoweredAI app on the device and provides the voice instruction for “Emotion Detection, which then causes the app to launch the camera. It continuously analyses the user’s face and the significance of the emotions detected by the server are converted into audio output and the app gives an immediate voice response to the user.

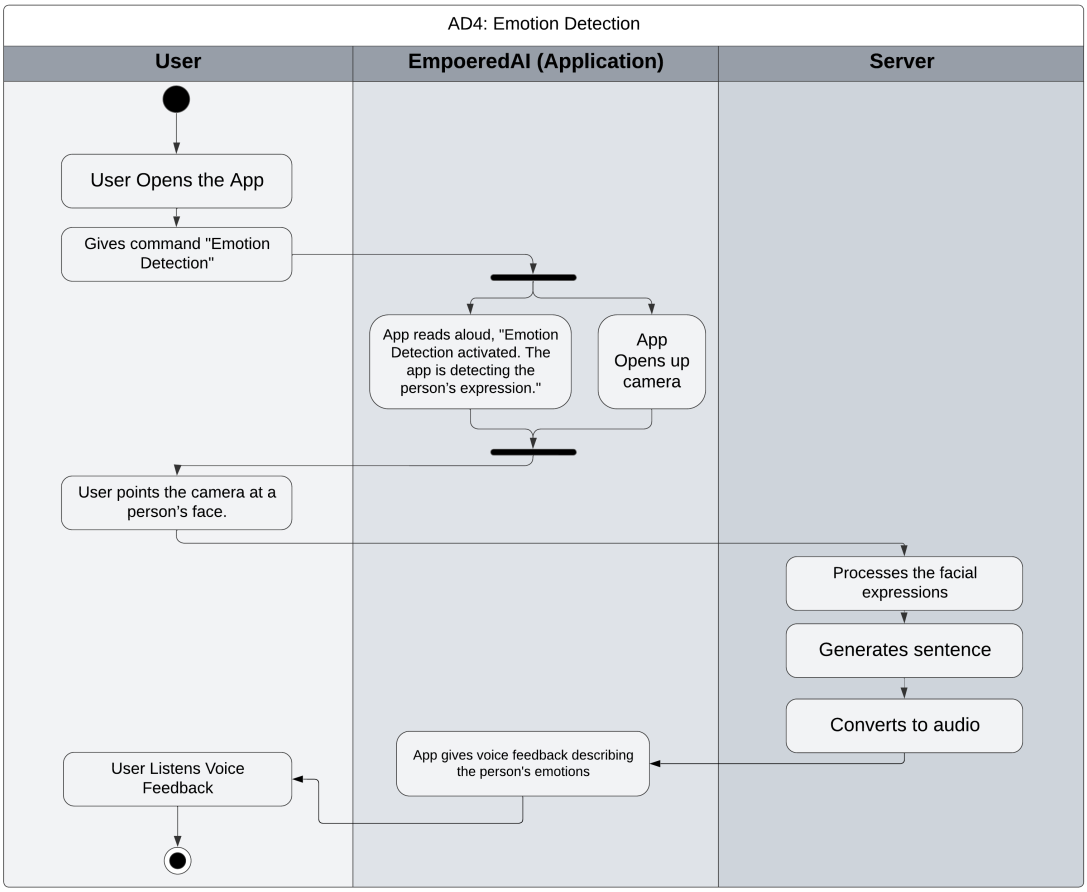

Figure 3.5: Activity Diagram 4 - Facial Emotion Detection

3.2.2 Class Diagram

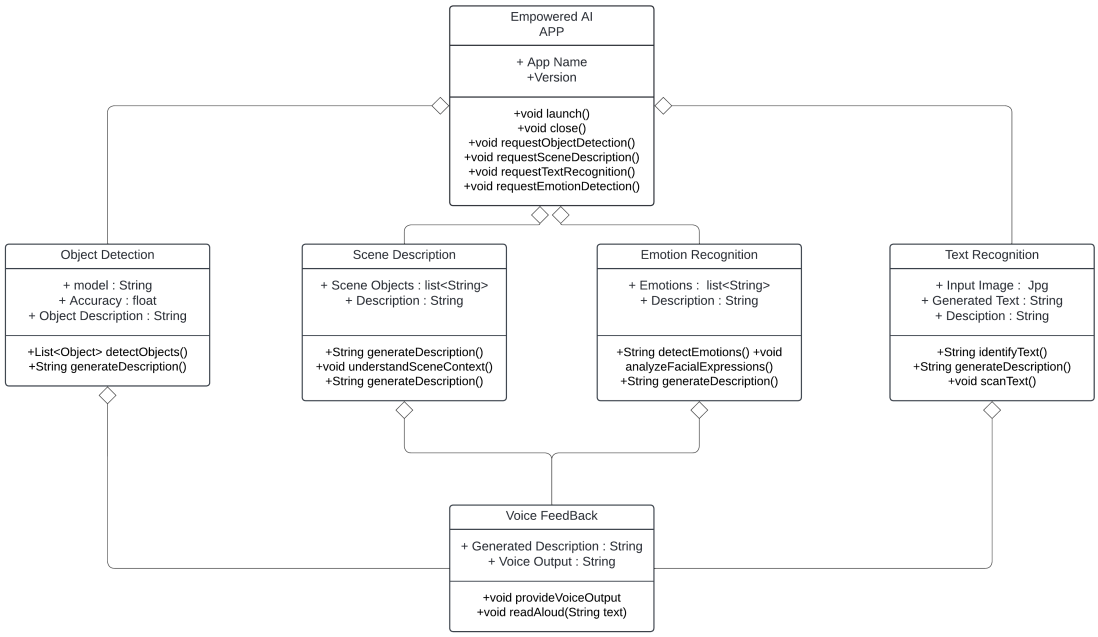

Figure 3.6: Class Diagram

The class diagram illustrates the main components of the Empowered AI App and how they interact to deliver functionalities like object detection, scene description, emotion recognition, text recognition, and voice feedback.

Empowered AI App Class: A master component of the system, the task of which is to receive users' demands and distribute them among individual subsystems.

Object Detection Class: Implements an object detection method based on a transferred model and provides textual descriptions of the objects detected in the scene.

Scene Description Class: Stores a list of the detected objects and creates an overall description of the scene based on certain context.

Emotion Recognition Class: Uses the particular scene to identify feelings and describe them using the facial expressions found.

Text Recognition Class: Takes a frame from a video and transcrites it into identifying the text contained in that particular frame.

Voice Feedback Class: Transforms generated description from different subsystems into voice for user to provide feedback.

Summary: The Empowered AI App is an object, scene, text and emotion recognition application with voice over for the user.

3.2.3 Class-Level Sequence Diagram

The sequence diagram illustrates the interaction between the user and various modules of the "Empowered-AI" app, from opening the app to processing different AI-powered functionalities.

Sequence Diagram Steps:

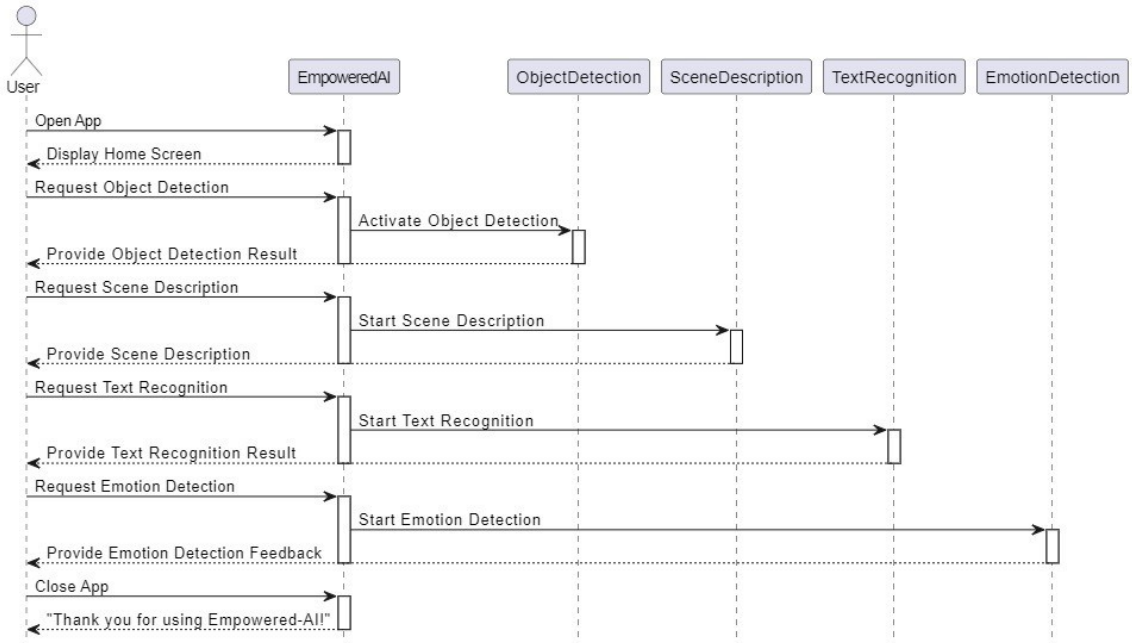

Figure 3.7: Class-Level Sequence Diagram

- **Open App:** The user opens the Empowered-AI app, which displays the home screen.
- **Request Object Detection:** The user requests object detection via a voice command, prompting the app to activate the ObjectDetection module to analyze the environment.
- **Provide Object Detection Result:** The app provides audio feedback, describing the detected objects.
- **Request Scene Description:** The user requests a scene description; the SceneDescription module processes and generates a description of the scene.
- **Provide Scene Description:** The app delivers the scene description as audio feedback to the user.
- **Request Text Recognition:** The user requests text recognition, triggering the TextRecognition module to scan and identify the text in the environment.

- **Provide Text Recognition Result:** The app reads the recognized text aloud to the user.
- **Request Emotion Detection:** The user requests facial emotion detection, and the EmotionDetection module identifies the emotions of individuals in the scene.
- **Provide Emotion Detection Feedback:** The app provides feedback on detected emotions via audio.
- **Close App:** The user closes the app, and a closing message is either displayed or spoken.

3.2.4 State Transition Diagram

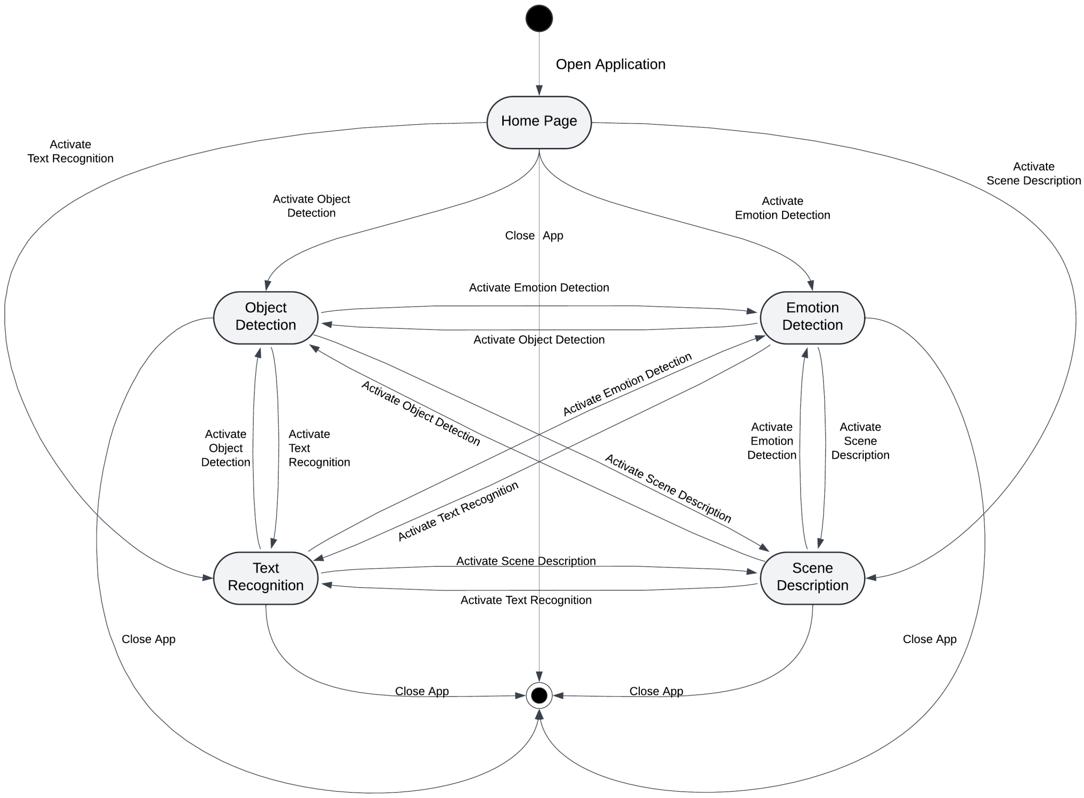

Figure 3.8: State Transition Diagram

The State Transition Diagram (Figure 3.8) illustrates the various states within the EmpoweredAI application and the transitions between them. The application begins at the "Home Page" state, from which the user can initiate various functionalities through corresponding voice commands. These include:

- **Object Detection:** Transition to this state occurs when the user activates object detection. Once object detection is completed, the user can either exit the application or activate another feature.

- **Emotion Detection:** This state is entered upon activating the emotion recognition feature. Similarly, the user can switch to other functionalities like object detection or scene description.
- **Text Recognition:** Upon activating text recognition, the system transitions to this state, allowing the app to recognize and process text from video frames.
- **Scene Description:** The user can also activate scene description from the home page or from other states, generating a description of the captured scene.

From each of these functional states (Object Detection, Emotion Detection, Text Recognition, Scene Description), the user can transition to other features or exit the application. Additionally, there is a direct transition to the **Close App** state from every functional state, allowing the user to exit the application at any point.

This diagram highlights the user's ability to seamlessly navigate between the various features of the EmpoweredAI app and the overall flow of interactions.

3.3 Data Design

The "Empowered-AI" app does not require any persistent storage or databases, as it focuses entirely on real-time processing. All data is processed temporarily in-memory and discarded after being used. Here's how the system handles data:

1. **Input Capture:**
   - **Visual Inputs:** Images or video frames are captured through the camera for object detection, scene descriptions, or emotion recognition.
   - **Text Inputs:** Optical Character Recognition (OCR) captures printed text.
   - **User Inputs:** Voice commands or taps trigger specific app functionalities.
2. **Processing:**
   - Data is processed in-memory using AI models for object detection, scene descriptions, and emotion recognition. Temporary data structures, such as lists or arrays, store intermediate results.
3. **Output Generation:**
   - Processed data is converted into voice feedback and immediately relayed to the user. The data is then discarded after use.

No databases are involved, as all data is managed in-memory for quick, real-time responses.